



Agenda

Distribution Key and Sort Key

D

Demo for Distribution Key and Sort Key

W

Visualize Redshift DWH

H

Use case for Data Lakes

Definition of Data Lakes

Data Lakes vs DWH

L

Services in AWS

A

(a) AWS S3

(d) AWS DynamoDB

K

(b) AWS Glue

E

(c) AWS Athena

Demo in AWS

Distribution Key and Sort Key

Distribution Key $\rightarrow$ [Key, All, Even]

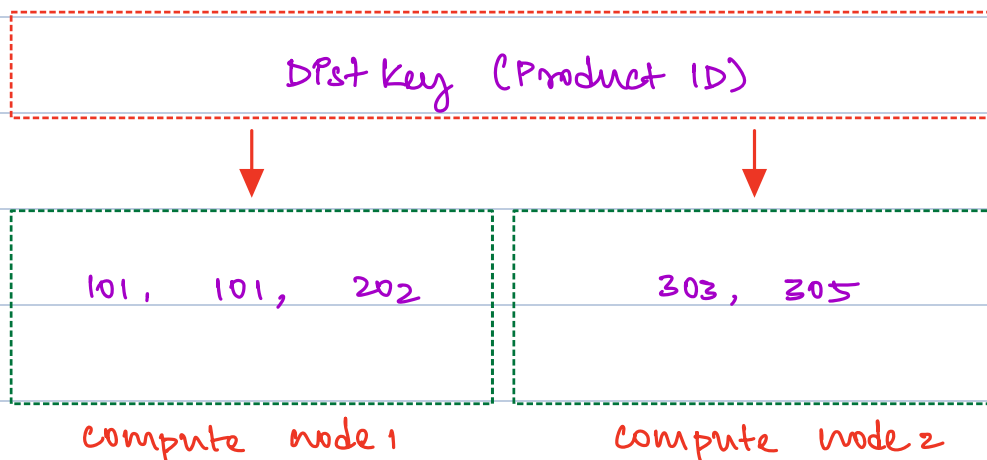
PKey	Product name	Price
101	Laptop	1900
101	Laptop	1900
202	Earphone	20
303	Mobile	1000
305	Keyboard	100

Product table

Distribution Key is the algorithm used by the leader node to distribute data to different worker nodes

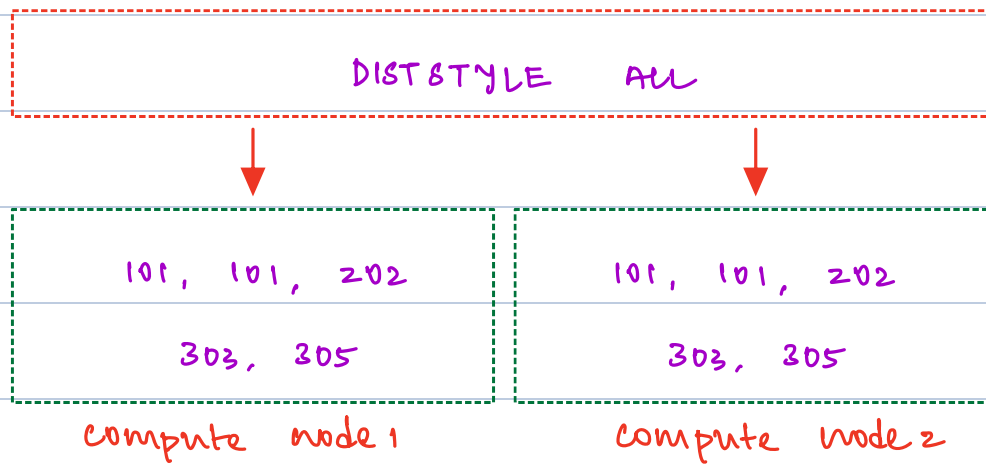
DstKey

Redshift uses a hashing algorithm on value in the DstKey column.



DIST style ALL

Complete table is stored in all compute nodes



DIST style EVEN

→ for tables where you don't frequently ^{Join}

Redshift distributes the rows across the worker in a simple round robin fashion



If you don't set a DIST style while creating the table it will be set as AUTO. While the table is small it will be ALL and as size increases it will shift to EVEN

Demo to Distribution Key and Sort Key



No Sort Key



Sort Key

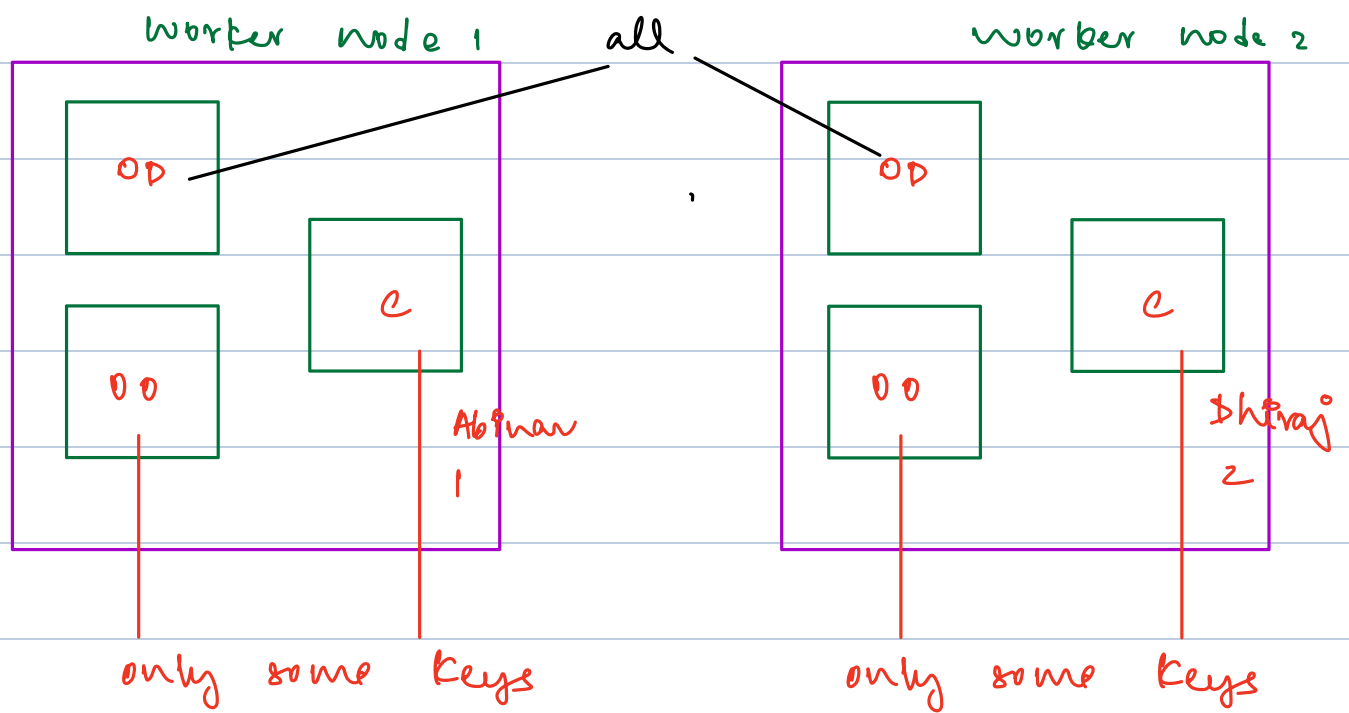
Visualize Data Warehouse

→ Gold Data

- ① Start with the use case
- ② Design the schema. Star vs Snowflake vs Galaxy
- ③ Run ETL and load the data inside your schema
- ④ Connect BI tools or run queries on top of your DWH

Clearly, data is of highest quality

and the data is consumed by Data Analyst and Business Analyst



1	2
2	3

orderId cust id

3	Dhiraj
2	Abhinav

cust id customer

Use case for Data Lakes

Data Warehouse is built for a purpose, for a single use case. However, what about Raw Data?

Netflix DWH → Hours of streaming

country-wise, show-wise, demographics

what about?

- Twitter tags reviewing a show
- user interaction, on device
- watch time of user for given show
- Youtube reviews of your show?



For you to store structured data, semi-structured data, unstructured data in Raw Format we use Data Lake

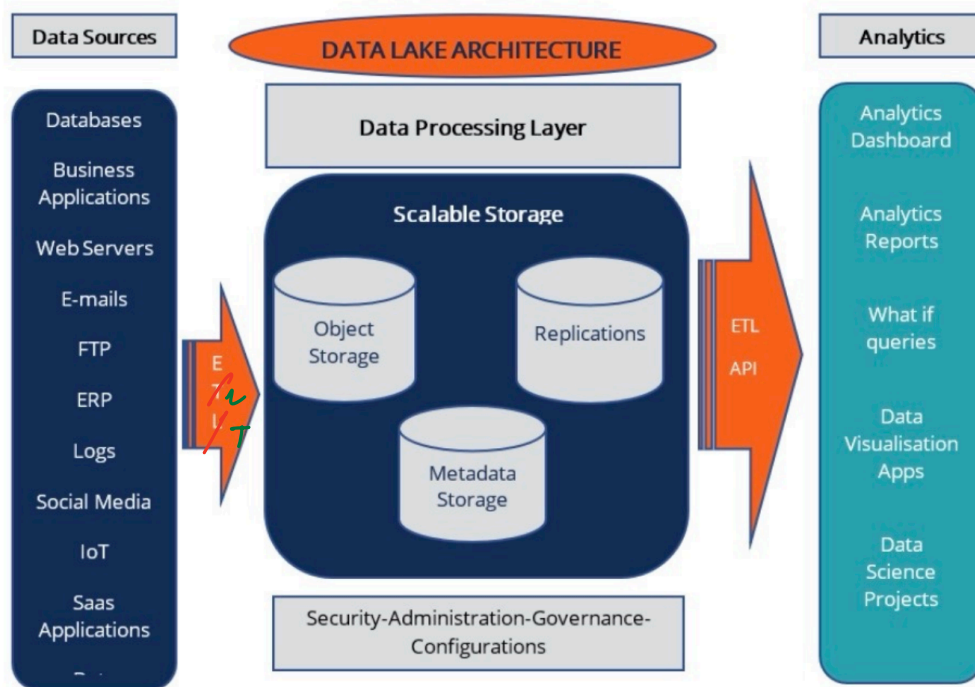
Data Lake

A Data Lake is a centralized repository designed to store, process, secure and analyse vast amounts of data in native Raw format.

Data Warehouse vs Data Lake

Data Lakes	Data Warehouse
① RAW Data Quality	① Golden Data Quality
② Usually in PB's	② Usually in TB's
③ Extract load Transform	③ Extract transform load
④ Extremely cheap	④ Usually expensive
⑤ Mostly used by Data Scientist	⑤ Mostly used by Data Analyst and Business Analyst

Data Lake Architecture



Tech stack

AWS : S3, DynamoDB, AWS Glue, AWS Athena

Amazon S3 → Object Storage in AWS.

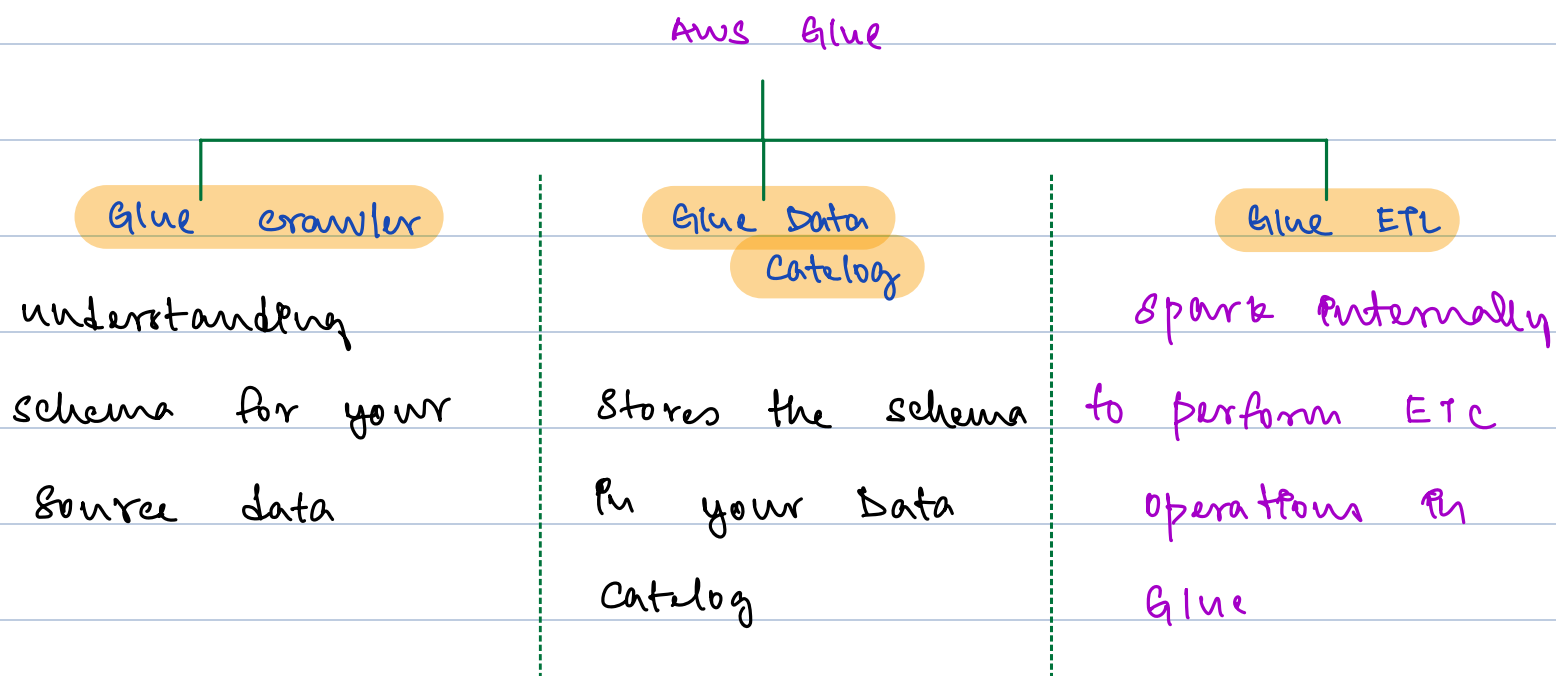
used for storing RAW data in our data lake.

SLA: 99.99999...

Amazon DynamoDB → NoSQL Database in the AWS

It is a fully managed No-SQL Key-Value and document database.

AWS Glue → Serverless data-integration and data processing service.

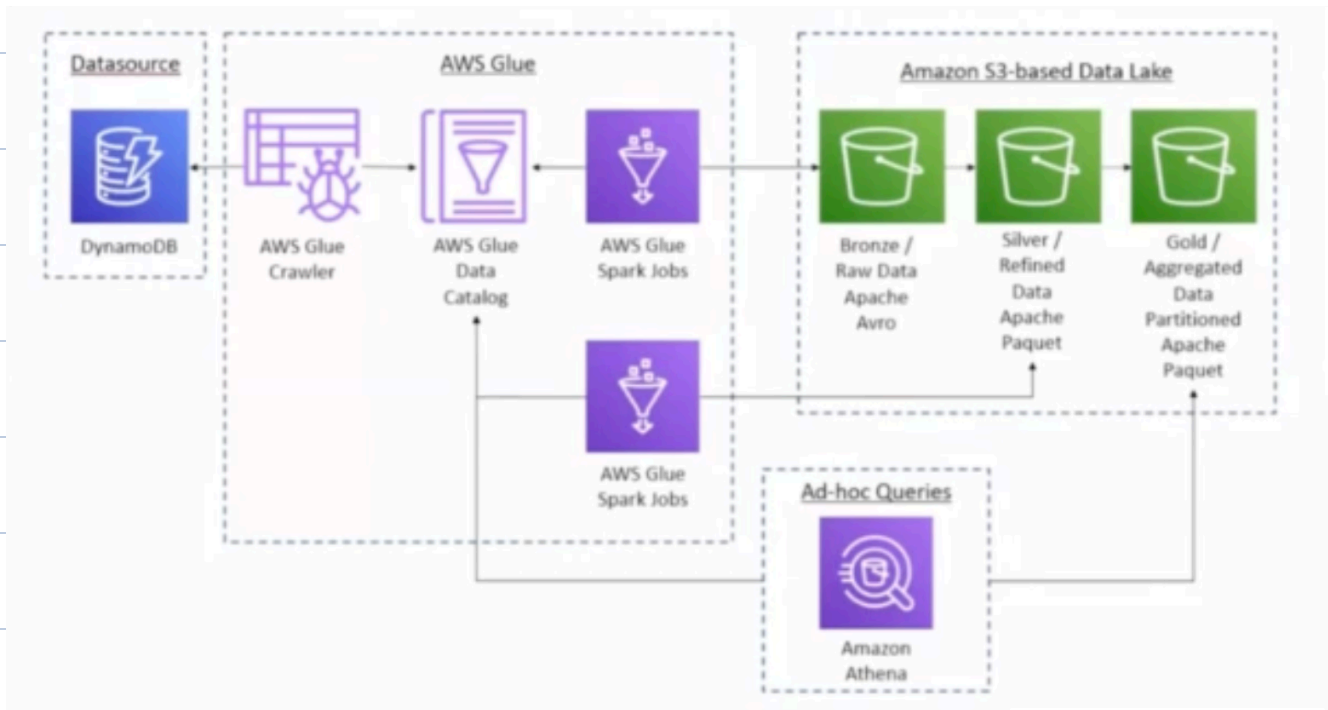


AWS Athena → Interactive query service that lets you analyze data directly in Amazon S3

Data Warehouse : schema on write

Data lake : schema on read

Demo In AWS



Next class

- ① AWS Glue ETL
- ② Athena for SQL Data Lakes
- ③ Lake Houses